

Pre-attentive processing

Pre-attentive

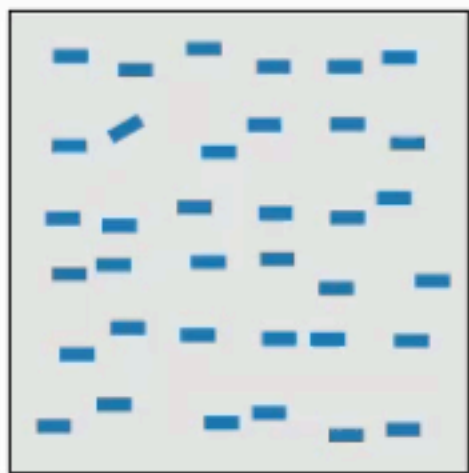
immediately recognize variation

with little or no conscious effort

(<200-250 ms)

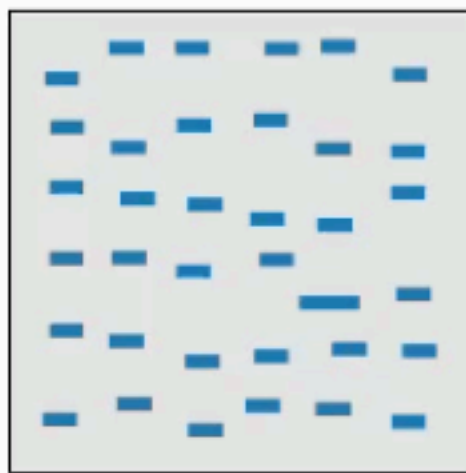
3

2



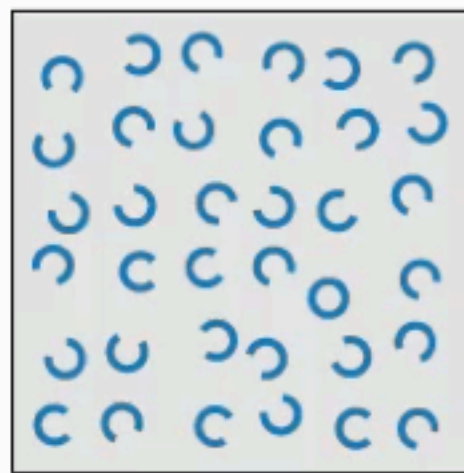
line (blob) orientation

Julész & Bergen 83; Sagi & Julész 85a, Wolfe et al. 92; Weigle et al. 2000



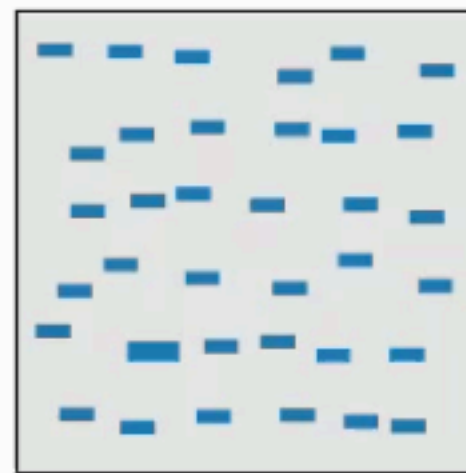
length, width

Sagi & Julész 85b; Treisman & Gormican 88



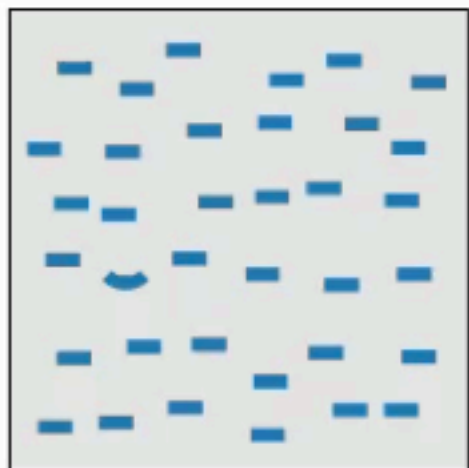
closure

Julész & Bergen 83



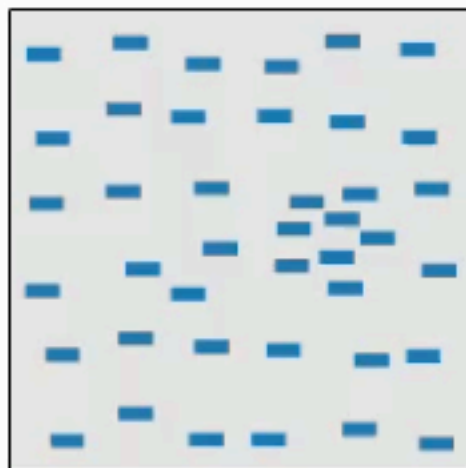
size

Treisman & Gelade 80; Healey & Enns 98; Healey & Enns 99



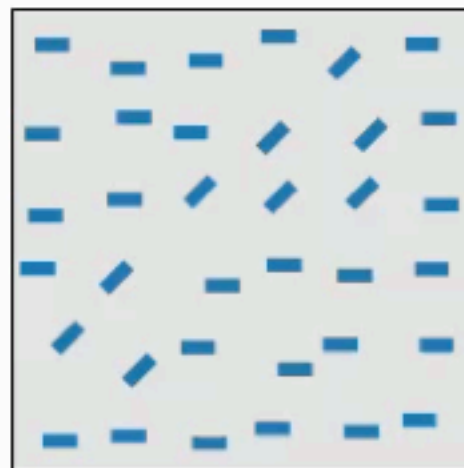
curvature

Treisman & Gormican 88



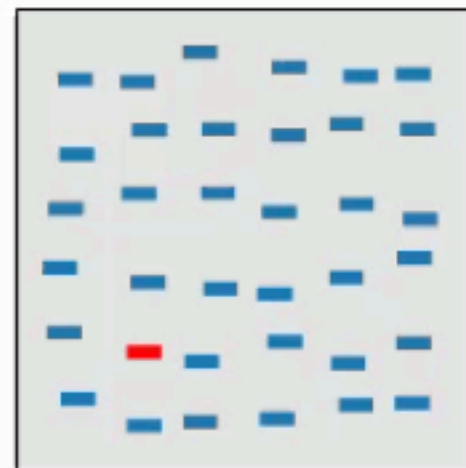
density, contrast

Healey & Enns 98; Healey & Enns 99



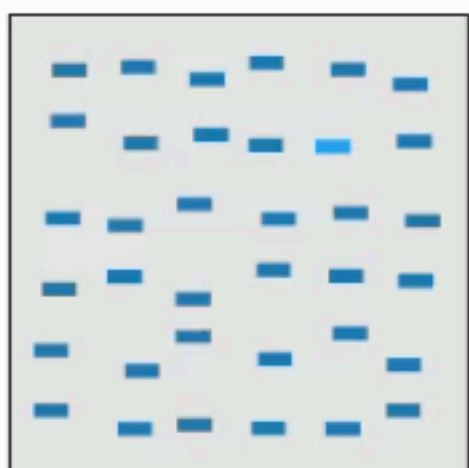
number, estimation

Sagi & Julész 85b; Healey et al. 93; Trick & Pylyshyn 94



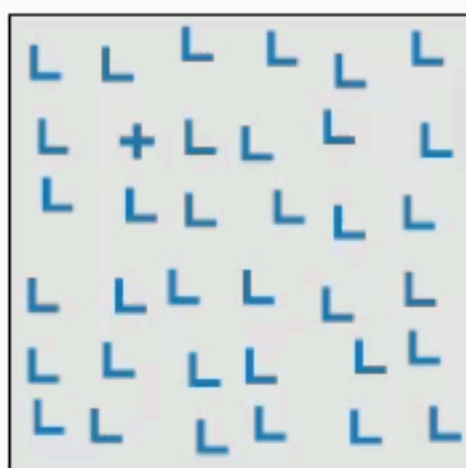
colour (hue)

Nagy & Sanchez 90; Nagy et al. 90; D'Zmura 91; Kawai et al. 95; Bauer et al. 96; Healey 96; Bauer et al. 98; Healey & Enns 99



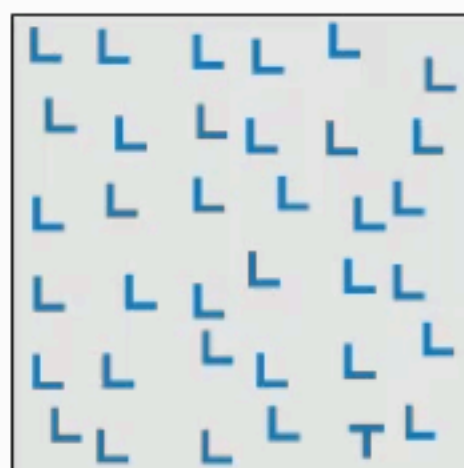
intensity, binocular lustre

Beck et al. 83; Treisman & Gormican 88; Wolfe & Franzel 88



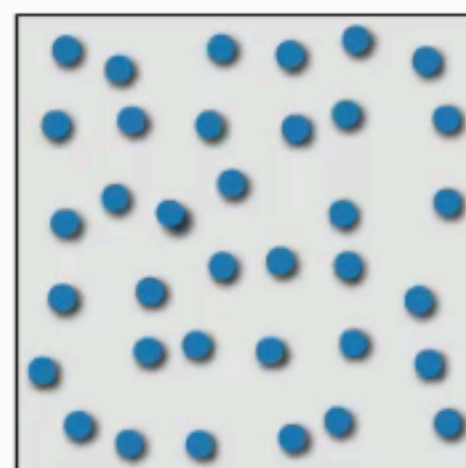
intersection

Julész & Bergen 83



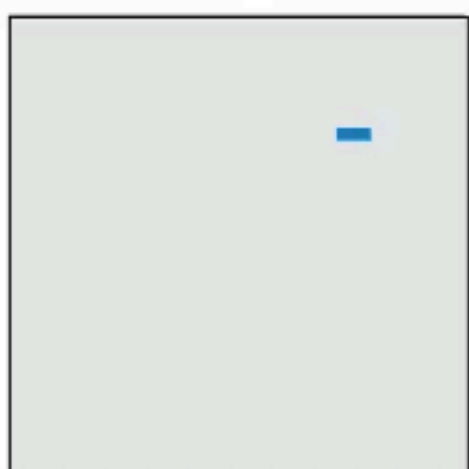
terminators

Julész & Bergen 83



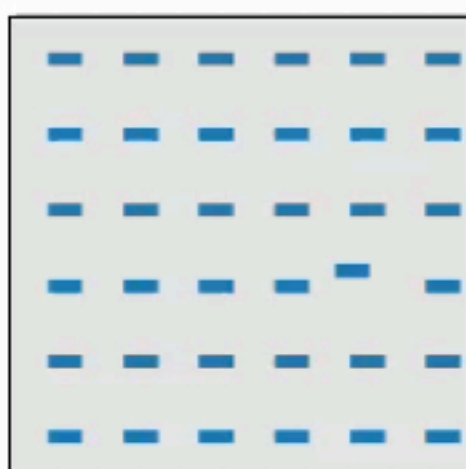
3D depth cues

Enns 90b; Nakayama & Silverman 86



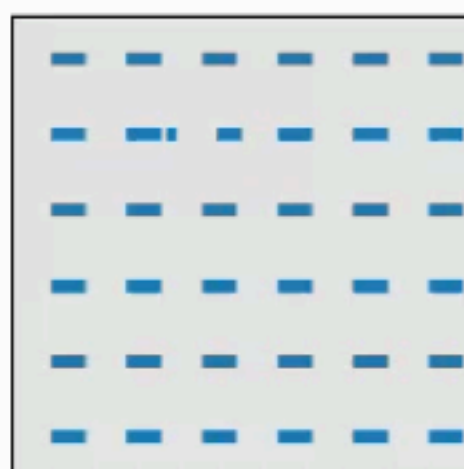
flicker

Gebb et al. 55; Mowbray & Gebhard 55; Brown 65; Julész 71; Huber & Healey 2005



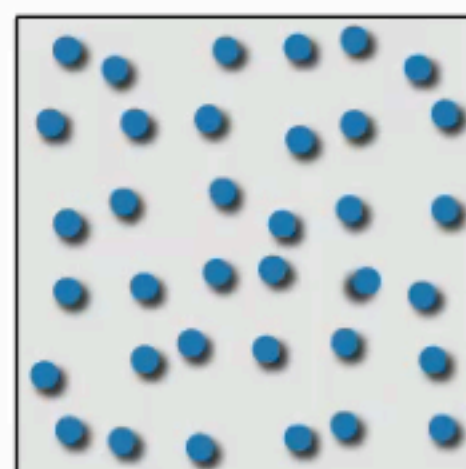
direction of motion

Nakayama & Silverman 86; Driver & McLeod 92; Huber & Healey 2005



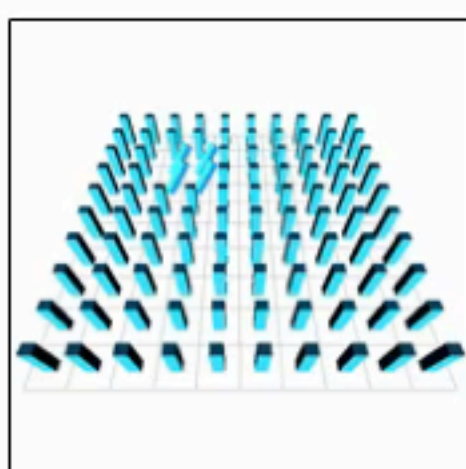
velocity of motion

Tynan & Sekuler 82; Nakayama & Silverman 86; Driver & McLeod 92; Hohnsbein & Mateeff 98; Huber & Healey 2005



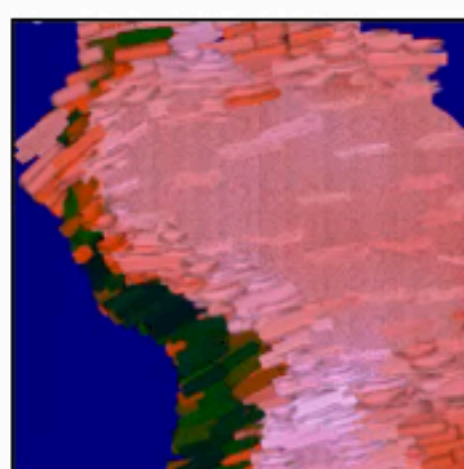
lighting direction

Enns 90a



3D orientation

Enns & Rensink 90a; Enns & Rensink 90b; Liu et al. 2003



artistic properties

Healey 2001; Healey & Enns 2002; Healey et al. 2004

Attentive

Takes some delicate effort to
perceive differences.

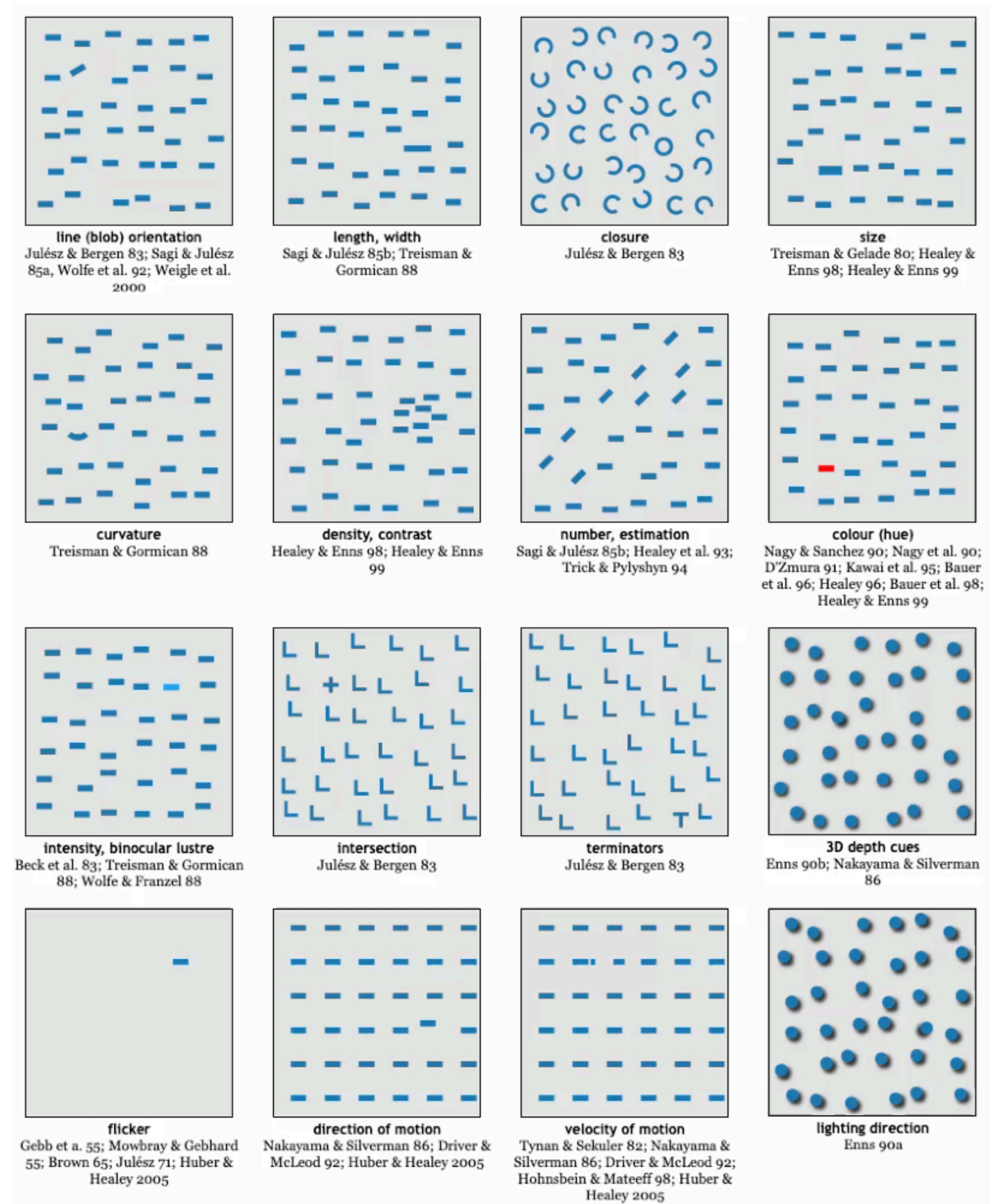
Pre-attentive processing

Pre-attentive

immediately recognize variation
with little or no conscious effort
($<200\text{--}250\text{ ms}$)

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Feature conjunctions: Is a **red circle** present?

